

Pim Protocol *Litepaper*

A concise summary of the Pim Protocol Technical Whitepaper: the problem, the architecture, token utility, security highlights, roadmap, and answers to the three questions readers ask most.

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The Problem

FINANCIAL EXCLUSION

Existing blockchain infrastructure assumes mid-range smartphones and reliable connectivity — excluding the populations most exposed to currency instability and stealth dollarisation [WP §3, §10.3].

MONETARY INSTABILITY

Prior energy-anchored designs relied on single-ratio minting formulas vulnerable to noisy or manipulated signals, and binary halt/no-halt circuit breakers with no graduated response [WP §6.1, §6.3].

The Whitepaper identifies seven compounding gaps in existing infrastructure spanning hardware accessibility, monetary policy design, quantum-cryptographic exposure, and privacy-versus-compliance trade-offs [WP §3].

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The Solution

Pim Protocol is a Layer-1 blockchain built on seven architectural pillars [WP §4.1]: an embedded ledger node architecture accessible on sub-2GB-RAM devices, PoEM-based hybrid consensus across 100 dynamic shards, a dual-token adaptive monetary system, full post-quantum cryptography from genesis, a hybrid VM scaling from microcontroller to GPU server, optional embedded-compatible privacy, and a fifth "Sovereign Edge Layer" hardware tier reaching \$3-class devices.

HIGH-LEVEL ARCHITECTURE

Application

PIM-VM Hybrid — Lite + Full modes, one runtime

Privacy

Cash Mode — optional, premium, embedded-compatible

Monetary ADTMS + EQCF + QolEx — graduated, demand-responsive
Cryptographic NIST post-quantum standards from genesis
Consensus PoEM + 100 DALI-optimised shards
Hardware \$3 Sovereign Edge Layer through GPU node

Full pillar detail: [\[WP §4.1, §5\]](#)

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Token Utility & Headline Tokenomics

TOKEN	TYPE	SUPPLY	UTILITY
QOL	Energy-backed flatcoin	Demand-responsive, no fixed cap	Payments, transaction fees, QolEx swaps
PYM	Fixed-supply governance token	100,000,000 (hard cap, genesis)	Governance, staking, PoEC bonding

QOL minting is calibrated by EQCF, a seven-stage pipeline weighting energy cost, hashrate, velocity, liquidity depth, and volatility into a bounded, persistence-gated scalar — designed so no single noisy signal can swing issuance [\[WP §6.1\]](#). PYM's cap is immutable; slashed PYM is routed to the SRP Buyback Reserve rather than destroyed, converting enforcement into a stability-funding mechanism [\[WP §6.2\]](#).

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Security Highlights

- ✓ Full NIST post-quantum cryptographic suite (FIPS 203/204/205 + HQC) deployed from the genesis block — nine years ahead of the 2035 transition deadline [\[WP §7\]](#)
- ✓ 51% attack requires simultaneous compromise of five structurally different node tiers across hardware, software, connectivity, and geography [\[WP §12.1\]](#)
- ✓ Fork probability of 0.0007% at 10,000 simulated nodes [\[WP §11\]](#)
- ✓ Eight identified protocol risk classes, each with a formally specified resolution — including signal-manipulation resistance requiring a 36-hour minimum sustained attack cost [\[WP §9\]](#)
- ✓ Deterministic execution enforced at the Shared Host API boundary — Lite Mode and Full Mode nodes produce byte-identical results [\[WP §9.1\]](#)
- ✓ Every EQCF cycle and ADTMS state transition is STARK-committed and ML-DSA signed on-chain, producing an independently verifiable monetary policy audit trail [\[WP §7\]](#)

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Roadmap

Q3 2026
Testnet — listed as an open engineering item in the Whitepaper [WP §8]
Mainnet
<i>Spec gap: no mainnet launch date specified in the Whitepaper.</i>
PimBridge

Cross-chain interoperability — referenced as "planned" for EU market integration; architecture not yet detailed [WP §10.2]

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Three-Question FAQ

Is QOL a stablecoin?

No. QOL is not pegged to any foreign currency or backed by off-chain reserves. Its stability comes from EQCF and ADTMS, which calibrate issuance against attested real network activity, targeting 1-2% volatility [WP §4.2, §6.1].

What happens if PYM is slashed for dishonest behaviour?

Slashed PYM is never burned. It routes entirely to the SRP Buyback Reserve, which defends QOL's peg — turning penalty enforcement into a source of monetary stability funding [WP §6.2, App. A].

Can a \$3 device really participate meaningfully?

Yes. The Sovereign Edge Layer specifies full coin sovereignty — independent verification of a device's own holdings — on 256KB-RAM hardware without a TPM, using PUF-based attestation and a lightweight local DALI model [WP §4.1 Pillar VII, App. B].

Notice. This Litepaper is a condensed summary of the Pim Protocol Technical Whitepaper, prepared by Pim Global Limited and Qollabs Ltd. Performance figures represent design targets; mainnet figures may vary. Pim Protocol has not yet reached mainnet launch. This document is not an offer to sell or a solicitation to buy any security or financial instrument, and is not investment advice.

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